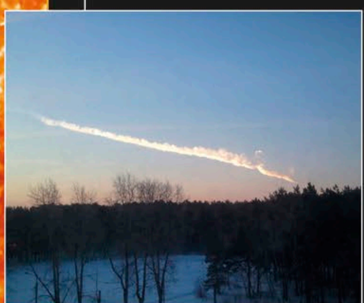
METEORS AND METEORITES

38–39 Gravity, motion, and orbits
170–177 Asteroids
216–219 Comets
Monthly sky guide 426–501

POPULARLY KNOWN AS SHOOTING STARS, meteors are linear trails of light-radiating material produced in Earth's upper atmosphere by the impact of often small, dusty fragments of comets or asteroids called meteoroids. About 1 million visible meteors are produced each day. If the meteor is not completely destroyed by the atmosphere, it will hit the ground and is then called a meteorite. If the meteorite is very large, a crater will be formed by the impact.

METEOROIDS

Most of the dusty meteoroids responsible for visual meteors come from the decaying surfaces of cometary nuclei. When a comet is close to the Sun, its surface becomes hot, and snow just below the surface is converted into gas. This gas escapes and breaks up the surface of the friable, dusty nucleus and blows small dust particles away from the comet. These dusty meteoroids have velocities that are slightly different from that of their parent comet. This causes them to have slightly different orbits, and as time passes, they form a stream of particles all around the original orbit of the comet. This stream is fed by new meteoroids every time the parent comet swings past the Sun. The inner solar system is full of these streams. Dense streams are produced by large comets that get close to the Sun. Streams with relatively few meteoroids are formed by smaller and more distant comets. As the Earth orbits the Sun, it continually passes in and out of these streams, colliding with some of the meteoroids that they contain. Names are given to some meteor showers that occur at fixed times of year, such as the Leonids (right).

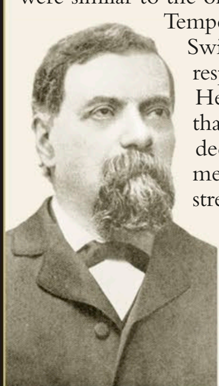


CHELYABINSK FIREBALL
This is the vapor trail left when a house-sized stony asteroid streaked across the predawn sky in Russia's southern Ural region on February 15, 2013. It broke apart before reaching the ground, but many fragments were recovered.

LEONID METEOR SHOWER
Leonid meteors are seen around November 17 every year and are so called because they appear to pour out of the constellation of Leo. Every 33 years, the shower strengthens into a veritable storm. The woodcut on the right was carved by the Swiss artist Karl Jauslin in 1888; it represents the maximum activity of the 1833 Leonids.

GIOVANNI SCHIAPARELLI

Giovanni Schiaparelli (1835–1910) was an Italian astronomer who worked at the Brera Observatory in Milan and has two claims to fame. In 1866, he calculated the orbits of the Leonid and Perseid meteoroids and realized that they were similar to the orbits of comets Tempel–Tuttle and Swift–Tuttle, respectively. He concluded that cometary decay produced meteoroid streams. In the late 1870s, he went on to map Mars's surface.

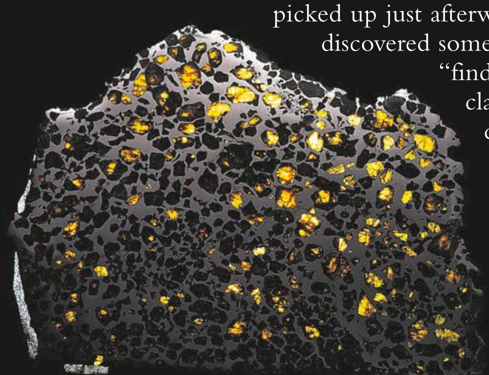


METEORITES

Small extraterrestrial bodies that hit the Earth's atmosphere are completely destroyed during the production of the associated meteor. If, however, the impacting body has a mass of between about 70 lb (30 kg) and 10,000 tons, only the surface layers are lost during atmospheric entry, and the atmosphere slows down the incoming body until it eventually reaches a "free-fall" velocity of just over 90 mph (150 kph). The central remnant then hits the ground. The fraction of the incoming body that survives depends on its initial velocity and composition. Meteorites are referred to as "falls" if they are seen to enter and are then picked up just afterward. Those that are discovered some time later are called "finds." Meteorites are classified as one of three compositional types.

STONY
This is by far the most common type of meteorite, comprising 93.3 percent of all falls. They are subdivided into chondrites and achondrites.

IRON
Iron meteorites make up 5.4 percent of all falls. They are composed mainly of iron-nickel alloy (consisting of 5–10 percent nickel by weight) and small amounts of other minerals.



STONY-IRON
The rarest meteorites—just 1.3 percent of meteorite falls—are a mixture of stone and iron-nickel alloy, similar to the composition of the rocky planets.